THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The period of life when a child becomes sexually mature is called: 1

(a) puberty **(b)** old age **(c)** childhood **(d)** infancy

2. Heat flows from a body at higher temperature to a body at: 1

(a) higher temperature **(b)** same temperature only
(c) absolute zero only **(d)** lower temperature

3. The SI unit of time is the: 1

(a) day **(b)** second **(c)** minute **(d)** hour

4. The process of taking food into the body is called: 1

(a) ingestion **(b)** egestion **(c)** digestion **(d)** absorption

5. Ovaries start releasing mature eggs at: 1

(a) puberty **(b)** birth **(c)** infancy **(d)** old age only

6. In conduction, the particles of the solid: 1

(a) fly off
(b) stop
(c) pass on heat to neighbours but stay in position
(d) move over long distances

7. Speed of an object is: 1

(a) distance travelled per unit time **(b)** total distance
(c) total time **(d)** time per unit distance

8. The finger-like projections in the small intestine are called: 1

(a) villi **(b)** nephrons **(c)** alveoli **(d)** cilia

9. Rapid growth of height at puberty is called: 1

(a) growth spurt **(b)** stagnation **(c)** gestation **(d)** decline

10. Dark-coloured clothes absorb ___ heat than light-coloured clothes. 1

(a) more **(b)** equal **(c)** no **(d)** less

11. The instrument in a vehicle which records the distance travelled is the: 1

(a) odometer **(b)** speedometer **(c)** barometer **(d)** thermometer

12. The exchange of gases between the body and the environment is: 1

(a) absorption **(b)** excretion **(c)** digestion **(d)** breathing

13. The age group of adolescence is roughly: 1

(a) 20 to 30 years **(b)** 1 to 5 years **(c)** 40 to 50 years **(d)** 11 to 19 years

14. The three modes of heat transfer are: 1

(a) radiation, freezing, boiling **(b)** convection, sublimation, condensation

(c) conduction, evaporation, melting **(d)** conduction, convection, radiation

15. 1 hour is equal to: 1

(a) 360 seconds **(b)** 60 seconds **(c)** 3600 seconds **(d)** 600 seconds

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is puberty? At what age does it begin? 2

17. Name the three modes of heat transfer. 2

18. Define speed and write its SI unit. 2

19. Name the five steps of nutrition in animals. 2

20. What is menstruation? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the changes in the body in adolescents under height, voice and hair. 3

22. Describe an activity to show conduction in a metal rod. 3

23. A pendulum completes 30 oscillations in 60 s. Find its time period and frequency. 3

24. Describe the path of food in the human digestive system. 3

OR

What is the role of adrenal gland? 3

25. Why are heaters placed near the floor and coolers/ACs near the ceiling? 3

OR

Convert: 72 km/h to m/s and 15 m/s to km/h. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the changes that occur during puberty in boys and girls. 5

OR

Explain conduction, convection and radiation with one activity or example each. 5

27. Describe the experiment to find the time period of a simple pendulum and show how it depends on length. 5

OR

Describe the human digestive system and the role of each organ with a labelled description. 5

END OF QUESTION PAPER — **Total: 50 Marks**